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Machine learning models to predict osteonecrosis in patients with femoral neck fractures undergoing internal fixation.

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CLINICAL & AGENTIC AI TAKEAWAY

The logistic regression model effectively predicts osteonecrosis of the femoral head in patients with femoral neck fractures after internal fixation, aiding in clinical decision-making.

ABSTRACT

--- REPOSITORY & PUBLISHER WEB METADATA --- Affiliations: Department of Orthopaedics, Peking University Third Hospital, 49 North Garden Rd, Haidian District, Beijing 100191, China; Engineering Research Center of Bone and Joint Precision Medicine, Ministry of Education, Peking University Third Hospital, 49 North Garden Rd, Haidian District, Beijing 100191, China. Keywords: Femoral neck fractures, Osteonecrosis, Machine Learning, Risk Prediction Extended Abstract from Repository: Objective This study aimed to use machine learning (ML) to establish risk factor and prediction models of osteonecrosis of the femoral head (ONFH) in patients with femoral neck fractures (FNFs) after internal fixation. Methods We retrospectively collected clinical data of patients with FNFs who were followed up for at least 2 years. Only intracapsular FNFs were included. In total, 437 patients and 24 variables were enrolled. The entire dataset was divided into training (89.5 %) and test (10.5 %) datasets. Six models-logistic regression, naive Bayes, decision tree, random forest, multilayer perceptron, and AdaBoost-were established and validated for predicting postoperative ONFH. We compared the area under the receiver operating characteristic curve (AUC), accuracy, recall, and F1 score of different models. In addition, a confusion matrix, density curve, and learning curve were used to evaluate the model performance. Results The logistic regression model performed best at predicting ONFH in patients with FNFs undergoing internal fixation surgery, with an AUC, accuracy, recall, F1 score, and prediction value of 0.84, 0.89, 1.00, 0.94, and 89.1 %, respectively. The learning and density curves demonstrated a good prediction fitting degree and distinct separation. When establishing the ML models, the reduction quality, internal fixation removal, American Society of Anesthesiologists classification, injury mechanism, and displacement distance of the medial cortex were the top five risk factors positively correlated with the occurrence of ONFH. Conclusions The logistic regression model had excellent performance in predicting ONFH in patients with FNFs after internal fixation and could provide valuable guidance in clinical decision-making. When choosing treatment options for patients with FNFs, doctors should identify the risk factors and consider using the presented models to help anticipate outcomes and select individualised treatment.

Keywords: Femoral neck fractures, Osteonecrosis, Machine Learning, Risk Prediction

Study Overview & Background

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Methodology & Clinical Focus

Study Classification: theoretical, modeling, or simulation study. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

The logistic regression model effectively predicts osteonecrosis of the femoral head in patients with femoral neck fractures after internal fixation, aiding in clinical decision-making.

Extended Repository Abstract

Objective This study aimed to use machine learning (ML) to establish risk factor and prediction models of osteonecrosis of the femoral head (ONFH) in patients with femoral neck fractures (FNFs) after internal fixation. Methods We retrospectively collected clinical data of patients with FNFs who were followed up for at least 2 years. Only intracapsular FNFs were included. In total, 437 patients and 24 variables were enrolled. The entire dataset was divided into training (89.5 %) and test (10.5 %) datasets. Six models-logistic regression, naive Bayes, decision tree, random forest, multilayer perceptron, and AdaBoost-were established and validated for predicting postoperative ONFH. We compared the area under the receiver operating characteristic curve (AUC), accuracy, recall, and F1 score of different models. In addition, a confusion matrix, density curve, and learning curve were used to evaluate the model performance. Results The logistic regression model performed best at predicting ONFH in patients with FNFs undergoing internal fixation surgery, with an AUC, accuracy, recall, F1 score, and prediction value of 0.84, 0.89, 1.00, 0.94, and 89.1 %, respectively. The learning and density curves demonstrated a good prediction fitting degree and distinct separation. When establishing the ML models,

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Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

Bibliographic Details & Indexing

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